

HEART SOUNDS

- **Definition** : Sounds produced by the mechanical activities of the heart during each cardiac cycle.
- **Causes of Production** :—
 - a. Flow of blood through cardiac chambers
 - b. Contraction of cardiac muscle
 - c. Closure of valves of the heart
- **Auscultation & Recording** : Heard using a **stethoscope** or **microphone**; graphical registration is studied via **phonocardiogram**.
- **Clinical Diagnostic Value** : Alterations in heart sounds indicate underlying cardiac pathologies (*especially valvular heart diseases*).

CLASSIFICATION OF HEART SOUNDS

- **1st & 2nd Heart Sounds** :—
 - Termed **classical heart sounds**.
 - Resemble the spoken words '**LUB**' (**LUBB**) and '**DUB**' (**DUP**) respectively.
- **3rd Heart Sound** :—
 - Mild sound; normally **not heard with a regular stethoscope**, but detectable using a sensitive microphone/phonocardiogram.
- **4th Heart Sound** :—
 - Inaudible physiological sound; **becomes audible only under pathological conditions**.
 - Studied primarily by phonocardiography.

COMPARATIVE TABLE OF HEART SOUNDS

Features	First Heart Sound (S ₁)	Second Heart Sound (S ₂)	Third Heart Sound (S ₃)	Fourth Heart Sound (S ₄)
Occurs during	Isometric contraction period	Protodiastole and part of	Rapid filling phase	Atrial systole

Features	First Heart Sound (S ₁)	Second Heart Sound (S ₂)	Third Heart Sound (S ₃)	Fourth Heart Sound (S ₄)
	and part of ejection period	isometric relaxation		
Characteristics	Long, soft, and low-pitched (Resembles 'LUBB')	Short, sharp, and high-pitched (Resembles 'DUP')	Low-pitched	Inaudible sound
Cause	Closure of atrioventricular (AV) valves	Closure of semilunar valves	Rushing of blood into ventricles	Contraction of atrial musculature
Duration (sec)	0.10 to 0.17	0.10 to 0.14	0.07 to 0.10	0.02 to 0.04
Frequency (Hz)	25 to 45	50	1 to 6	1 to 4
Relation with ECG	Coincides with peak of 'R' wave	Precedes or appears 0.09 sec after peak of 'T' wave	Between 'T' wave and 'P' wave	Between 'P' wave and 'Q' wave
Vibrations on Phonocardiogram	9 to 13	4 to 6	1 to 4	1 to 2

APPLIED PHYSIOLOGY OF HEART SOUNDS

- **1. Reduplication / Splitting of Heart Sounds :—**
 - Means splitting of normal heart sounds.
 - **S₁ Splitting** : Occurs when atrioventricular valves (mitral and tricuspid) do not close simultaneously.
 - **S₂ Splitting** : Occurs when semilunar valves (aortic and pulmonary) do not close simultaneously.
- **2. Soft Heart Sounds :—**
 - Heart sounds become soft when intensity is ↓ decreased.

- Characteristically heard in **low blood pressure (hypotension)** and shock.

- **3. Loud Heart Sounds :—**

- Heart sounds produced with much higher intensity than normal (*e.g., hyperdynamic states, hypertension, valvular stenosis*).